

- 1 During a wet weather, a stationary rain droplet falls from great height and eventually reaches a terminal velocity of 9.00 m s^{-1} .

(a) Explain how the droplet achieved terminal velocity.

- As the droplet initially falls from rest, its velocity increases/accelerates downwards due to net downward force of weight.
 - Drag force due to air resistance acts upwards and it increases with increasing velocity.
 - So the net downward force (and hence by N2L, its acceleration) decreases until zero.
- The droplet will then reach a constant velocity. [3]

- (b) The droplet eventually hits the side window of a moving train. The relative velocity, v of the droplet makes an angle of θ with the vertical, as shown in Fig. 1.1.

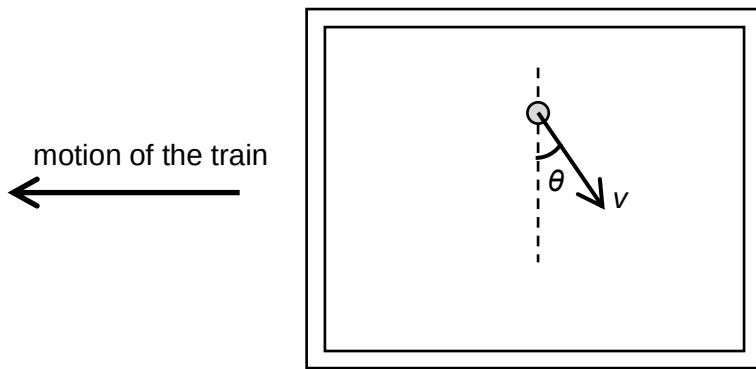


Fig. 1.1

Sketch a graph of the train speed against θ using the grid in Fig. 1.2.

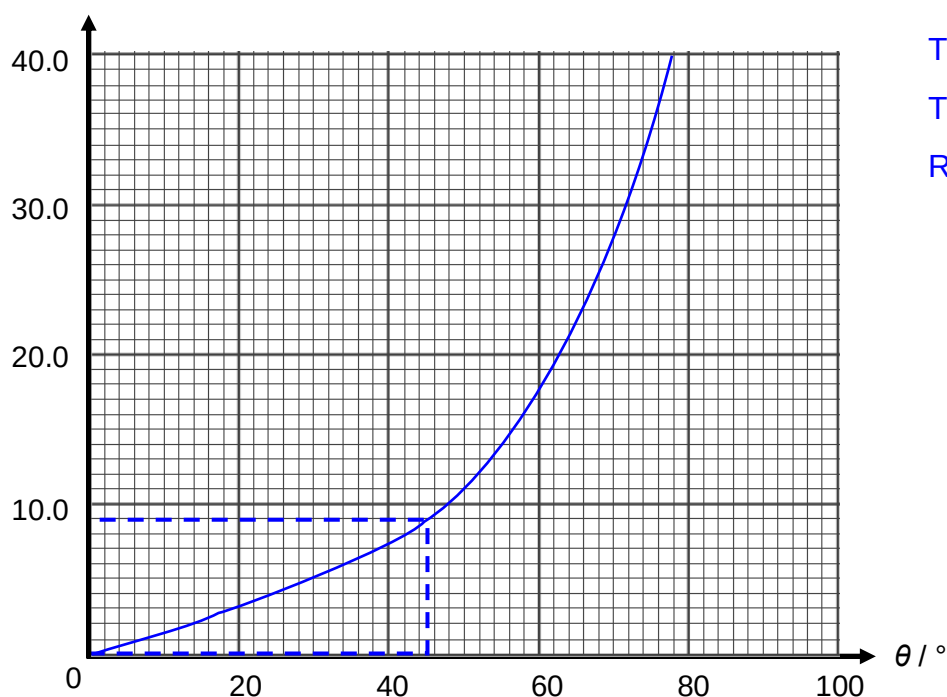
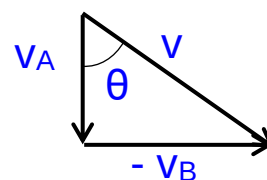
train speed / m s⁻¹

Fig. 1.2

Terminal velocity of droplet = v_A Train speed = v_B Relative velocity $v = v_A - v_B$ 

$$\tan \theta = v_B / v_A = v_B / 9$$

$$\therefore \text{train speed } v_B = 9 \tan \theta$$

[3]

- (c) The train is travelling at a speed of 125 km h⁻¹. On reaching the station, the brake is applied such that the train decelerates uniformly. During the deceleration, the train travels another 850 m before coming to a rest.

Determine the time taken for the train to stop.

$$\begin{aligned}
 s &= 0.5(u + v)t \\
 t &= 2s / (u + v) \\
 &= 2(850) / (125 / 3.6 + 0) \\
 &= 49 \text{ s}
 \end{aligned}$$

time taken = s [2]

[Total: 8]

